Modelization of migration flows and stocks disaggregated at the place of birth in the framework of non homogeneous markov chains

Mathieu Canovas

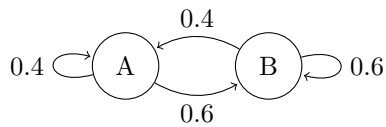
July 2023

1 Methodology: Markov Chains with non homogeneous transition probabilities

1.1 Markov Chain Refresher

Formal definition with discrete state space $(X_n)_{n\in\mathbf{N}} \sim$ Markov(λ, P)
where P is a n $\times$ n transition probability i.e. a stochastic by row matrix, where we have at row i transition probabilities from i to j for $j \in [1, ..., n]$. When P does not depend on the time n, we say that the Markov chain is homogeneous. Markov chains were introduced by Andrey Markov in 1906. (Markov (1906), Behrends (2000)).
The particularity of such a sequence is the lack of memory. The probability to reach another state depends only on the current state.

Graph Representation



The associated transition probability is then:

$$\mathbf{P} = \begin{array}{c} \\ A \\ B \end{array} \begin{array}{c} \begin{array}{cc} A & B \end{array} \\ \begin{pmatrix} 0.4 & 0.6 \\ 0.4 & 0.6 \end{pmatrix} \end{array}$$

In this particular case, it is easy to see that the probability to reach any state does not depend on the current position. Therefore, whatever the initial distribution, we obtain immediately with λ P the invariant distribution.
The invariant distribution is defined such that $\lambda\ P = \lambda$ where λ is some vector of initial distribution probability giving information on where are we more likely to start in this state space.

1.2 Modelization of migration flows

1.2.1 Defining Transition Probabilities time and birth-place specific

Let us look at a transition probability time and birth-place specific with three state space A,B and C. Set the birth place k = A and t = t_1.

$$\mathbf{P}^{\mathbf{A}}_{\mathbf{t_1}} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} P_{t_1}(Z^A_{A,A}) & P_{t_1}(E^A_{A,B}) & P_{t_1}(E^A_{A,C}) \\ P_{t_1}(R^A_{B,A}) & P_{t_1}(Z^A_{B,B}) & P_{t_1}(T^A_{B,C}) \\ P_{t_1}(R^A_{C,A}) & P_{t_1}(T^A_{C,B}) & P_{t_1}(Z^A_{C,C}) \end{pmatrix} \end{array}$$

which can be rewritten this way:

$$\mathbf{P}^{\mathbf{A}}_{\mathbf{t_1}} = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{pmatrix} P_{t_1}(X^A_{t_2} = A | X^A_{t_1} = A) & P_{t_1}(X^A_{t_2} = B | X^A_{t_1} = A) & P_{t_1}(X^A_{t_2} = C | X^A_{t_1} = A) \\ P_{t_1}(X^A_{t_2} = A | X^A_{t_1} = B) & P_{t_1}(X^A_{t_2} = B | X^A_{t_1} = B) & P_{t_1}(X^A_{t_2} = C | X^A_{t_1} = B) \\ P_{t_1}(X^A_{t_2} = A | X^A_{t_1} = C) & P_{t_1}(X^A_{t_2} = B | X^A_{t_1} = C) & P_{t_1}(X^A_{t_2} = C | X^A_{t_1} = C) \end{pmatrix} \end{array}$$

where $(X^A_{t_n})_{n\in\mathbf{N}}$ is Markov(λ^A, $(P^A_{t_n})_{n\in\mathbf{N}}$). We get back the return, emigration and transit flows Azose and Raftery (2019) and the stayers Abel et al. (2019).
Now, the idea to get back the stocks and the birth place specific bilateral migration flows to then compute the differences in stocks and assess the error of estimation depending on the decision rule you apply to the value given at the cell (i,j) of the delta, follows these steps:

1.2.2 Procedure to obtain theoretical stocks, flows and estimations

: We proceed for k = A:

Definining initial distributions λ_A birth-place specific.

Based on transition probabilities and initial distribution, compute the distribution by place of residence across time Compute the distribution across the system A,B,C of individuals born in A - assuming no death, no birth between this interval of time, from t_0 to t_1 and assuming we keep track of the same individuals in this interval of time. In other words, compute $LX^A_{t_1} := \lambda^A P^A_{t_1}$.
And from, t_0 to t_n, one can simulate the evolution of the distribution of people born in A across countries A,B and C.

A picture of that, for the example of four countries and three periods of transitions:

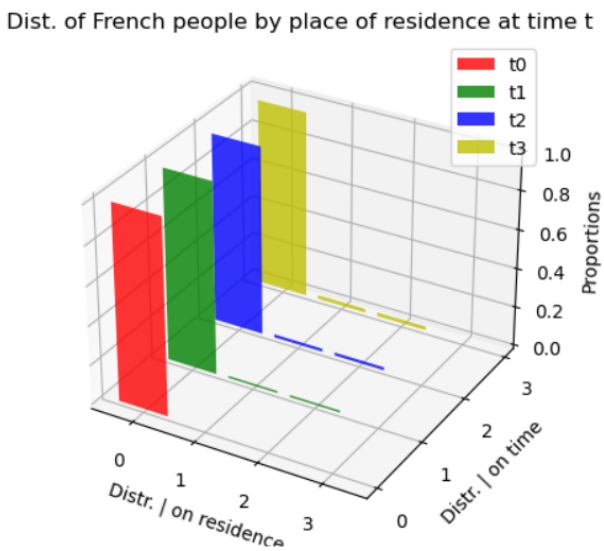


Figure 1: 0-France, 1-Morocco, 2-Turkya, 3-Syria

This graph shows us that all the individuals born in France started in France at t0. Then, between t0 and t1, up to t3, due to the nature of the time and French-born specific transition probabilities, we obtain the following slices of distributions in space from t0 to 3.

Based on the distributions, the initial distribution and the total population, compute the stocks Compute the stocks by setting an initial number at time t_0 of individuals for the sub-population of interest $Pop^A_{t_0}$ and deduce the stocks from the margins previously computed i.e. $(LX^A_{t_n})_{n\in\mathbf{N}}$.
$S^A_{t_n} := LX^A_{t_n} \times Pop^A_{t_0}\ \forall n \in \mathbf{N}$.

Based on stocks and transition probabilities, compute the flows We deduce the birth place specific bilateral migration flow with this formula: $F^A_{t_n} := \text{diag}(S^{A,A}_{t_1}, S^{A,B}_{t_1}, S^{A,C}_{t_1}) P^A_{t_1}$.

Compute differences in stocks Assume n periods, then compute $\forall k \in [1, ..., n]$:
$\Delta S^A_{t_k,t_{k-1}} = S^A_{t_k}$ - $S^A_{t_{k-1}}$

Compute a decision rule to interpret the difference in stocks For the case of our system of three countries A,B,C:
Φ from $R^3 \times R^3$ into $R^3 \times R^9$ where it assigns for any cell of the matrix $\mathbf{M}$ in R^9 a composition of three quantities in the simplex generated by considering Transit T^A, Emigration E^A and Return flows R^A.
It gives, for example with k = 2,

$\Phi(\Delta S^A_{t_2,t_1}) = (Prop^{ij}_{E^A}, Prop^{ij}_{T^A}, Prop^{ij}_{R^A})_{ij \in [A,B,C]\times[A,B,C]}$.
where $Prop^{ij}_{E^A} + Prop^{ij}_{R^A} + Prop^{ij}_{T^A} = 1$ and each of these quantities are in between 0 and 1.
For simpler notations:

$\Phi_{ij}(\Delta S^A_{t_2,t_1}) := (Prop^{ij}_{E^A}, Prop^{ij}_{R^A}, Prop^{ij}_{T^A})$.

In the case of drop negative:

$$\Phi^{DN}_{ij}(\Delta S^A_{t_2,t_1}) = \begin{cases} \begin{cases} \emptyset & \text{if } i{=}j \\ \begin{cases} (1,0,0) & \text{if } (\Delta S^A_{t_2,t_1})_{ij} > 0 \\ \emptyset & \text{if } (\Delta S^A_{t_2,t_1})_{ij} < 0 \end{cases} & \text{otherwise} \end{cases} \end{cases} \tag{1}$$

In the case of reverse negative:

$$\Phi^{RN}_{ij}(\Delta S^A_{t_2,t_1}) = \begin{cases} \begin{cases} \emptyset & \text{if } i{=}j \\ \begin{cases} (1,0,0) & \text{if } (\Delta S^A_{t_2,t_1})_{ij} > 0 \\ (0,1,0) & \text{if } (\Delta S^A_{t_2,t_1})_{ij} < 0 \end{cases} & \text{otherwise} \end{cases} \end{cases} \tag{2}$$

$$\Phi^{+D,RN}_{ij}(\Delta S^A_{t_2,t_1}) = \begin{cases} \begin{cases} \begin{cases} (1,0,0) & \text{if } (\Delta S^A_{t_2,t_1})_{ij} > 0 \\ (0,1,0) & \text{if } (\Delta S^A_{t_2,t_1})_{ij} < 0 \end{cases} & \text{if } i{=}j \\ \begin{cases} (1,0,0) & \text{if } (\Delta S^A_{t_2,t_1})_{ij} > 0 \\ (0,1,0) & \text{if } (\Delta S^A_{t_2,t_1})_{ij} < 0 \end{cases} & \text{otherwise} \end{cases} \end{cases} \tag{3}$$

In the case of Abel's methodology:

EN CHANTIER

$$\Phi^{Abel'sInterpretation}_{ij}(\Delta S_{t_2,t_1}) = \begin{cases} \begin{cases} \emptyset & \text{if } i{=}j \\ \begin{cases} (1,0,0) & \text{if } (\Delta S^A_{t_2,t_1})_{ij} > 0 \\ (0,1,0) & \text{if } (\Delta S^A_{t_2,t_1})_{ij} < 0 \end{cases} & \text{otherwise} \end{cases} \end{cases} \tag{4}$$

Φ^{DN} : = Decision rule assuming emigration flows for positive differences off the diagonal; Beine et al. (2011)
Φ^{RN} : Decision rule assuming emigration flows for positive differences and return flows for negative difference off the diagonal Beine and Parsons (2015).
$\Phi^{+D,RN}$: = Decision rule using diagonal elements and using reverse negative;

where $\emptyset$:= No decision taken (will lead to zero). <u>Remark</u>: Notice that the decision rule relative to Reverse Negative is the same as the one induced by Abel's approach by filling bilateral migrant flow tables conditional on place of birth with a negative variation in stocks in the home country and only positive variations in stocks in host countries or a positive variation in stocks in the home country and only negative variations in stocks in host

countries.
This contingency table method enables, with migrant minimization assumption, to define a decision on how to interpret negative differences and positive differences depending on the sign of the variations in stocks off the diagonal.

If the variation is positive in home country, due to migrant minimization, it saturates and fill with zero entries the off-diagonal values of the row indexed by the home country's name.

Now the problem would be to improve the decisions we take concerning the interpretation of off diagonal values of our difference method.

One can remark that running simulations indeed permit to distinguish between cases or scenarios in order to be able to identify which decision rule would be optimal.

The problem is, among scenarios s_1,...,s_N choose the optimal decision rule Φ such that it reduces the error of estimation $E_t := \sum_i F_t^i$ (obtained at step 4) - $\mathbf{H}_t$ where H_t is the estimation function depending on the decision rule Φ (defined at step 6) that is function of $\Delta S_{t_1,t_2}$ (obtained at step 3).

Then find the optimal $\Phi^*_{s_i}$ for each scenario s_1,...,s_N considered.

In other words, solve: $\underset{\Phi}{argmin} \sum_k ||F_{t_k}(s_i) - H_{t_k}(\Delta S_{t_k,t_{k-1}(S_{t_k},S_{t_{k-1}})(s_i)}, \Phi)||_F$

where s_i can be quantified via the sequence of transition probabilities we define $\forall$ k sub-population considered of our system via $(P^k_{t_n})_{n\in\mathbf{N}}$.

1.3 A little procedure to characterize our markov chains and ask for an invariant distribution

1.3.1 Irreducibility

Reminder A markov chain is irreducible if there is a unique communicating class.
A communicating class is a subset of states such that each couple of states inside of it has a strictly positive probability to move from one to another at a given time of the discrete sequence.

For example, with scenario 1, the last transition probability still characterized by war is defining one closed communicating class consisting of A,B and C (since D excluded do not receive any emigration flows from A,B,C neither transit or return flows by people born in D from these countries A,B and C) and one open class consisting of one state D (since people from D might still want to run away from the war).

1.3.2 Positive Recurrent

1.3.3 Aperiodic

1.4 Interpretation of the dimensions of our simulation

1.4.1 Time Dimension

Our Markov Chains are modeled via non homogeneous transition probabilities and specific to each sub-population conditional on place of birth.

The chain $(X^k_{t_0},\ X^k_{t_1},\ X^k_{t_2},\ X^k_{t_3})$ represents the evolution of the position of a given individual born in k starting at $X^k_{t_0}$ and ending at $X^k_{t_3}$.

Here time plays the role of a stopping times defined as constant time variables to relate with the data available. To be more precise, $(X^k_{t_0},\ X^k_{t_1},\ X^k_{t_2},\ X^k_{t_3})$ could be in fact $(X^k_t)_t$ in product with $(\mathbf{I}(t)_{t\in\{t_0,t_1,t_2,t_3\}})_t$. Therefore, we do not model all the flows resulting from the difference in positions between t_k and t_{k+1} for every k in $(0,1,2)$.

Now the interpretation we want to have to relate with the available migrant stock data provided by the United Nations, is that t_{k+1} - $t_k = 5$ for every k in $(0,1,2)$ with years as units.

Therefore, the flows $F_{t_k}^{ij}$ will be equal to $\sum_l \sum_{i=1}^{n_l} \mathbf{I}_{\{X_{t_k-1}^{l,i}=i, X_{t_k}^{l,i}=j\}}$ where l correspond to place of birth's index and n_l total population born in l. In other words, whatever the path taken by individuals to go from i to j between t_{k-1} and t_k, we will only ask if between this interval of time, individual i born in k moved from i to j. This is to relate with how the data is collected by the UN and use the simulation results to provide recommendations in the use of certain methods rather than others depending on the case we have.

For example, we will have a difference between a short-term war and a long-term war. The short-term war will last for less than a period while the long-term war will last for more than a period. This would have consequences in terms of which probabilities to put in terms of return and emigration flows in direction of this country.

1.4.2 Spatial Dimension

This is for later improvements of the model. We could take into account distances between each place of residence that would impact the cost of migration.

Nevertheless, here we have spatial entities that are in a non-metric space, we do not consider any physical distance in our model to make it simpler. These are only different states one can be at.

1.4.3 Decision-Making Dimension

By Markov property, the transition probabilities and the current state of each simulated individual determine the random process of which next state this individual will be at.

We do not make endogeneous any form of rational thinking. Here we rather model collective action that may be irrational. This could be the object of more thinking. For example, when war occurs at a given country, we do not simulate any form a decision making depending on the level of wealth of each individual, the age and so on. We simply consider that people will emigrate massively and not return till the war has ended. Now what does this model ? An aggregate phenomenon related to a sum of rational decision making or is it rather a collective action due to the natural instinct of survival and fear of death?

From the OECD (`https://doi.org/10.1787/migr_outlook-2008-en`), we will try configure scenarios of different return probabilities depending on the country of destination and of place of birth.

1.4.4 Migration Comprehension

The flows we will simulate are purely related to differences in terms of place of residence and not related at all to the duration of stay after any movement. This could be controlled via a continuous time markov chain or via a discrete time markov chain with additional periods. We could simulate 15 years and try to quantify migration flows rigorously by adding the duration criterion (period of stay after movement from previous place of residence).

Interpretation of the migration flows Define $\mathbf{P}_k(X_{n+1} = i | X_n = j)$ as the probability to move from j to i for an individual born in k between n and n+1 i.e. in a given interval of time equal to five years.
It means in particular that we will have people that directly emigrated from j to i in this interval of time as well as people who moved from j to l in between this interval of time and eventually moved to i.
This probability, as the flows that will result from it, corresponds to an aggregation of direct/indirect transitions from j to i.

Emigration Probabilities Draženović et al. (2018)

Return Probabilities The main findings of the part III of the report of the OECD entitled "International Migration Outlook 2008" is focused on the description of return flows.
Notice first that, our definition of return flows will not necessarily match the one of the OECD in its most elementary approach as "the situation where a migrant goes back to his home country after living in another country for some period of time". Take for instance an individual moving between t_1 and t_2 from i to j, born in j, but living in

j between t_1 and t_2 - ϵ. Therefore, this individual did went back to his home country j after living in i for an epsilon period. We could define this as an ϵ-return flow. Since we do not simulate this in our model (which could be done nevertheless via a continous time markov chain), we do not quantify this in our model. We simply consider return flows, whatever epsilon between 0 and 5, where epsilon represent years.
We could also investigate the content of these articles Hagan and Thomas Wassink (2020) (economic sociology of return migration), Byron and Condon (1996) (historical context based analysis of return phenomenon) that could help us improve the understanding of this phenomenon and help us distinguish relevant cases to emphasize how the estimation methods perform well or not depending on the context.

Transit Probabilities These transit flows are also called by the literature "Secondary Migration" (OECD report of 2008 - International Migration Outlook).

1.4.5 Macro socio-economic Dimension

Here we would like to list key macro socio-economic characteristics to have differences between each nodes as well as macro socio-economic clusters and shocks to set a socio-economic background and some scenario's shocks.

- Human Development Index (HDI); Global Domestic Product (GDP);
- State of War;
- Economic Crisis;
- Natural Disaster;

We would like to emphasize that, optimally, it would have been interesting to try to fit a model to historical and well quantified migration flows phenomena. Nevertheless, this is not what we will do here and rather put en emphasis on the differences between the transition probabilities. For example, with an economic crisis, we will increase the probability of emigrate from less developed countries to more developed countries. To be more specific, we only take into account the sign of the impact of one shock on a particular transition probability heuristically according to intuition or imperfectly fact-based rationality but we do not use any tool to quantify this based on historical facts or theory.

1.4.6 How many periods should we simulate for each scenario ?

If we only want to simulate flows function of difference in positions in time with without taking into account duration of stay, we fix three periods to cover 15 years.

If we want to simulate flows by considering difference in positions as well as duration of stay, we could take 15 periods and then make a difference between temporary-permanent emigration flows as well as return flows as defined by the OECD (from host to home with period of stay in host of at least one year) and then, after 5 periods, computing the estimation and the bias. This would improve the analysis by taking into account short-term cross flows that intersect with two different periods of five years, the first one being between t0 and t4.

1.5 First Scenario

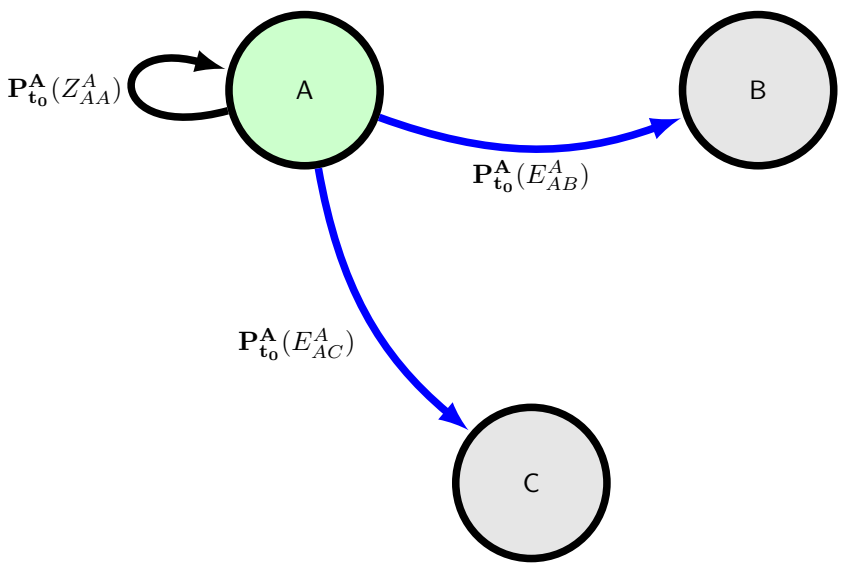
1.5.1 Shocks and General Logic

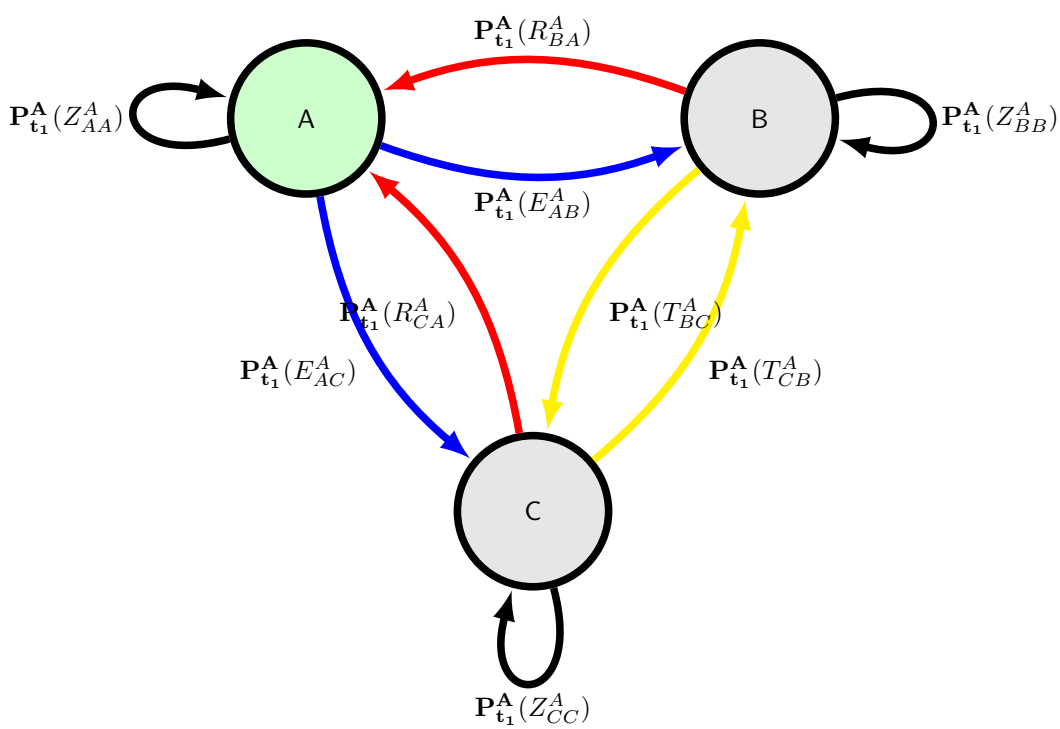
Initial Setting Few transits, small probability to emigrate, probability of stay to host country slightly higher than to return to home country (it means we do not consider that countries have different capabilities to keep immigrants and that in general a large part of emigrants return home after a five year period of time).

Shocks, Effects on Transition Probabilities

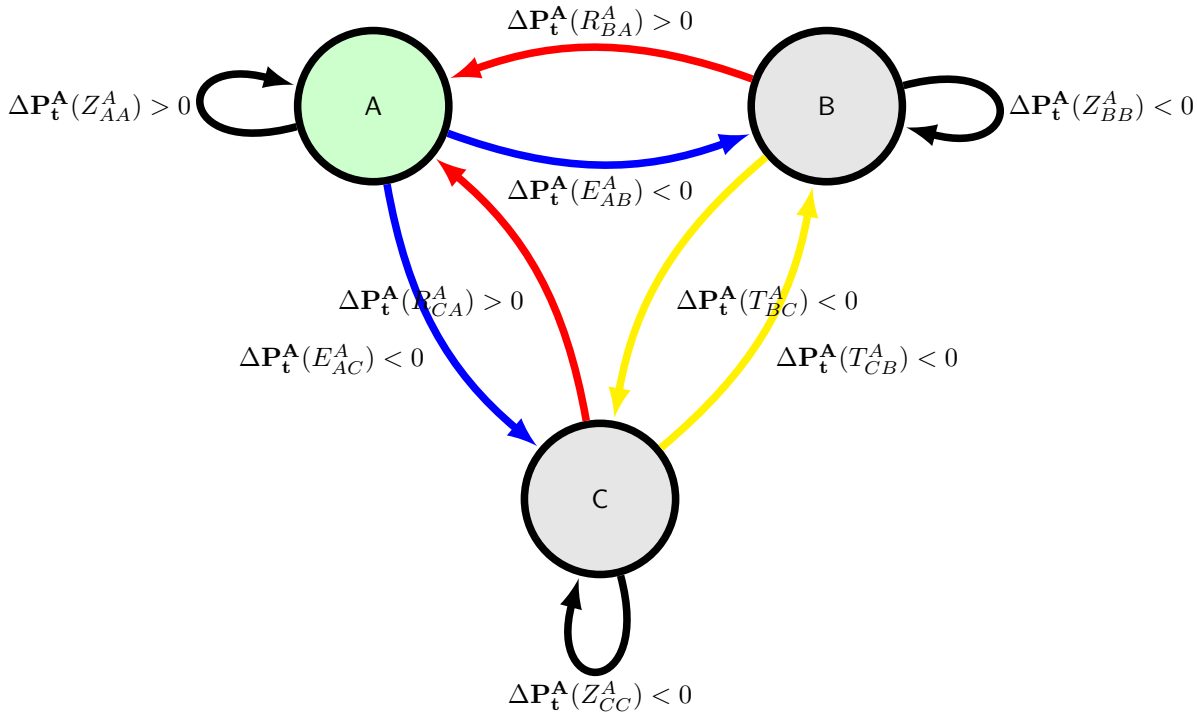
- War Shock at t1 and lasting two periods i.e. at least 10 years; (setting return flows/emigration flows to 0);
- Movements of return from A to B and to C between t2 and t3 other a five year interval of time; (increasing return flows)

t0-t1/t1-t2 dynamic We look at how will be modeled the dynamic of the population born in A between t0 and t2 in scenario 1, see in annex the code and parameters: Fixing country k = A :





(t2-t3 dynamic) We modify the transition probabilities according to a specific shock in order to emphasize a limit or an advantage of the difference method Considering a war sparked between the two entities B and C, we could change the probabilities according to this logic:



1.6 Manufacture of other scenarios

1.6.1 Ideas related to Return Flows

Based on the OECD report of 2008, part III : Let us underline that this collect of information on the dynamics of return, emigration and transit flows among a specified group of destination/origin countries is only to have some facts to back choices we can make concerning the transition probabilities i.e. to emphasize that they were not chosen out of nowhere or at least not in contradiction with statistics provided by international/national statistical institutions.

Information 1: The OECD informs that "departures by foreigners from OECD countries can represent anywhere between 20% and 70% of arrivals in any given year". This could mean that emigration and return transition probabilities can be very close.
Proposition 1:: In order to use directly the information provided by the OECD, we could simulate over 15 years and discriminate between temporary migration and long term migration.
On the feasibility: We would still have the same procedure to compute the error terms. What would change is that we could quantify all missing temporary migration flows related to temporary migration flows.

Information 2: Distinguishing return rates depending on the age.
We are given the information that, **depending on the life cycle of migrants**, there are difference in return rates. There are higher rates for the young and for retirees.
Proposition 2: Economic Crisis, High Unemployment - High Emigration Rates from young population, positive variation in the rate of emigration to richer countries. After a five year period, increasing return rates according to this U-curve logic.

Information 3: Comparison of the level of development between country of origin and country of destination.
"As the income disparities are greater, migrants are more likely to stay put." In particular, "Return rates to OECD countries are on average twice as high as those to developing countries."
Proposition 3: Setting greater return rates for people from OECD countries living in foreign OECD countries.
Setting lower return rates for people not from OECD countries living in OECD countries.

Information 4: One country may collectively decide via representative democracy to vote naturalisation or portability of social entitlements. As the report emphasizes, this can impact the duration of stay of migrants and therefore increase potentially their probability to stay or equivalently decrease the probability to return or to proceed to "secondary migration" i.e. transit migration.

Proposition 4: We can think of scenarios where countries increase naturalisation or decide to enforce the portability of social entitlements and as a consequence modify the transition probabilities for at least a period of five years, if not until the end of the simulation - assuming political stability.

Information 5: Political, economic and social in the home country is stable and attractive, return is only an option.

Proposition 5: Set a floor-return probability bearing this stability and attractiveness characteristics of the host country i.e. the probability of return related to this option calculus based on the agent decision making.

Final Remark We recommend the curious reader to look at the report of the OECD if desiring to study how return migration is measured and analysed with real data.

1.6.2 Ideas related to Emigration Flows

Relating to proposition 4 of previous subsection, one can think of increasing emigration rates due to internal/external-politically related shocks with, as external, economic shocks via a new technology, a prosperous economic context and, as internal, via new policies that are implemented to keep economic migrants for instance.

1.6.3 Ideas related to Transit Flows

1.7 Analysis of the errors

Errors relative to the difference method The error term can be decomposed as, in the case of reverse negative (containing the case of drop negative):

$$\hat{F}_{ij} - F_{ij} = ((\Delta S^{ij}_{t+1})_+ + (\Delta S^{ji}_{t+1})_-) - (E^i_{ij} - R^j_{ij} - \sum_{k \neq i,j} T^k_{ij})$$

which is equivalent to this:

$$\hat{F}_{ij} - F_{ij} = (\hat{E}^i_{ij} - E^i_{ij}) + (\hat{R}^j_{ij} - R^j_{ij}) - \sum_{k \neq i,j} T^k_{ij}$$

In the case where $\hat{E}^i_{ij} = 0$ and $\hat{R}^j_{ij} = 0$, we have:

$$\hat{F}_{ij} - F_{ij} = -E^i_{ij} - R^j_{ij} - \sum_{k \neq i,j} T^k_{ij}$$

In the case where $\hat{E}^i_{ij} > 0$ and $\hat{R}^j_{ij} = 0$, we have:

$$\hat{F}_{ij} - F_{ij} = -R^i_{ji} - R^j_{ij} - \sum_{k \neq i,j} (T^i_{jk} - T^i_{kj}) - \sum_{k \neq i,j} T^k_{ij}$$

Then obtain $\sum_{k \neq i,j}(T^i_{jk} - T^i_{kj})$ by taking $e'(F^i_t)_j$ - $e'(F^i_t)^T_j$ - E_{ij} - R_{ji} and T^k_{ij} via $(F^k_t)_{ij}$ In the case where $\hat{E}^i_{ij} = 0$ and $\hat{R}^j_{ij} > 0$, we have:

$$\hat{F}_{ij} - F_{ij} = -E^i_{ij} - E^j_{ji} - \sum_{k \neq i,j} (T^j_{ki} - T^j_{ik}) - \sum_{k \neq i,j} T^k_{ij}$$

Then obtain $\sum_{k \neq i,j}(T^j_{ki} - T^j_{ik})$ by taking $e'(F^j_t)^T_i$ - $e'(F^j_t)_i$ - E_{ji} - R_{ij} and T^k_{ij} via $(F^k_t)_{ij}$ In the case where $\hat{E}^i_{ij} > 0$ and $\hat{R}^j_{ij} > 0$, we have:

$$\hat{F}_{ij} - F_{ij} = -R^i_{ji} - \sum_{k \neq i,j} (T^i_{jk} - T^i_{kj}) - E^j_{ji} - \sum_{k \neq i,j} (T^j_{ki} - T^j_{ik}) - \sum_{k \neq i,j} T^k_{ij}$$

We suggest to use these previous calculus in order to compute a prediction model of the emigration and return flows.

Errors relative to the migration rate method Here we decomposed the error term in terms of transit, emigration and return flows.

$$\hat{F}_{ij}^t - F_{ij}^t = \frac{S_t^{ij}}{\sum_{l,l' \neq i,j} S_t^{l,l'}} \times M - E_{ij}^{i,t} - R_{ij}^{j,t} - \sum_{k \neq i,j} T_{ij}^{k,t}$$

where M $= \sum_{k \in I}(TR_{t+1}^k - TR_t^k)_+ = \sum_{k \in I}(\sum_{j \in I} S_{t+1}^{jk} - S_t^{jk})_+ = \sum_{k \in I}(\sum_{j \in I} E_{jk}^{j,t} + \sum_{l \neq j,k} T_{lk}^{j,t} - R_{kj}^{j,t} - \sum_{l \neq j,k} T_{kl}^{j,t})_+$.
Notice that with this M we avoid the case where, with only emigration flows in same quantities from all the countries of the system to their neighborhood country, we estimate zero total net flows.
where $\sum_{l,l'|l \neq l'} S_t^{l,l'} = \sum_{l,l'|l \neq l'} (Z_{l'l'}^l + R_{l'l}^{l,t} + \sum_{k \neq l,l'} T_{l'k}^l)$.

Notice that:

$$\hat{F}_{ij}^t - F_{ij}^t = \frac{Z_{jj}^{i,t-1} + E_{ij}^{i,t-1} + \sum_{k \neq i,j} T_{kj}^{i,t-1}}{\sum_{l,l' \neq i,j} S_t^{l,l'}} \times M - E_{ij}^{i,t} - R_{ij}^{j,t} - \sum_{k \neq i,j} T_{ij}^{k,t}$$

$$= \frac{\sum_{n=1}^{t-1}(Z_{jj}^{i,t-n} + E_{ij}^{i,t-n} + \sum_{k \neq i,j} T_{kj}^{i,t-n})}{\sum_{l,l' \neq i,j} S_t^{l,l'}} \times M - E_{ij}^{i,t} - R_{ij}^{j,t} - \sum_{k \neq i,j} T_{ij}^{k,t}$$

As we said during the presentation of this method, one can see that the assumption is that $R_{ij}^{j,t}$ and $T_{ij}^{k,t}$ are equal to 0 and that $Z_{jj}^{t-1} = 0$ (current foreign stocks explained by direct previous emigration flows) and that the next flows are explained according to the same dynamic. Indeed, under these assumptions:

One has that $\hat{F}_{ij}^t - F_{ij}^t = \frac{E_{ij}^{i,t-1}}{\sum_{l,l'|l \neq l'} E_{l,l'}^{l,t-1}} \times \sum_{k \in I} \sum_{j \in I} E_{jk}^{j,t}$ - $E_{ij}^{i,t}$.
Such that

$$\hat{F}_{ij}^t = \mathbf{E}(E_{ij}^{i,t}), E_{ij}^{i,t} \sim B(\sum_{k \in I} \sum_{j \in I} E_{jk}^{j,t}, \frac{E_{ij}^{i,t-1}}{\sum_{l,l'|l \neq l'} E_{l,l'}^{l,t-1}}).$$

Such that, under these assumptions, the expected quadratic error $\mathbf{E}((\hat{F}_{ij}^t - F_{ij}^t)^2)$ would be equal to the variance of E_{ij}^t which is easy to compute.

Nevertheless, in reality, with transit and return flows, the error term is much more complex to analyze and decompose. After a certain period of time, the previous stocks are furthermore the result of numerous waves of anterior flows, contrary to this assumption underlying this estimation method.

Eventually, we can see $\hat{F}_{ij}^t$ as an estimation of emigration flows and we can also see it as a structurally biased estimation method since it does not estimate return neither transit flows. Therefore we have a first point of comparison with difference methods where emigration and return flows are estimated.
As we can anticipate, we should have an over-estimation of emigration flows with most of its estimates.

We will try to decompose what is at the source of this over-estimation i.e. to what extent is it related to the estimated total net flows M and to the ratio of foreign-stock over the sum of all foreign-stocks.

Error relative to the poisson regression with migrant minimization approach of Abel 2013 We look at the error term relative to the estimation of migration flows proposed by Abel in his paper of 2013 consisting in defining as many bilateral migrant flow table as there are place of birth and filling them with a strong assumption on the stayers along with an estimation of off-diagonal terms via MLE.

There are 81 cases. But we can simplify the implementation with conditional matrix in Python via numpy and its function where.

How to implement the computation of this method and the computation of the associated error term ?

1) **Computing the estimations following contingency table methods**: Proceed to Migrant Minimization on each birth place specific table, fill them with the parameters obtained via Gradient Descent or proceed to method 1 with some tricks since we do simulations on 4 countries' cases.

$\Rightarrow$ Compute the estimated bilateral migrant flow table with migrant minimization assumption:
For each k, define $\hat{F}_t^k$, set the margin constraints.

Proceed to migrant minimization with a function and as inputs the migrant stock tables of time t and t-1. Return the birth place specific bilateral migrant flow table. Next step is to fill it.

<u>Method 1</u>: Using some tricks

The setting:
We want to obtain this: $(\hat{F}_t^A,\ \hat{F}_t^B,\ \hat{F}_t^C,\ \hat{F}_t^D)$ with constraints on the margins given by, respectively for each place of birth and for row margin/column margin $(((S_t^{Aj})_j, (S_{t+1}^{Aj})_j), ((S_t^{Bj})_j, (S_{t+1}^{Bj})_j), ((S_t^{Cj})_j, (S_{t+1}^{Cj})_j), ((S_t^{Dj})_j, (S_{t+1}^{Dj})_j)$ and constraints on the diagonal with migrant minimization.
After the migrant minimization step, we have the following information given the place of birth k :

$\phi((\Delta S_{t+1}^{kj})) = (I_{(\Delta S_{t+1}^{kA})_+>0},\ I_{(\Delta S_{t+1}^{kB})_+>0},\ I_{(\Delta S_{t+1}^{kC})_+>0},\ I_{(\Delta S_{t+1}^{kD})_+>0})$.
It returns precious information related to which cells we will then compute.
If $I_{(\Delta S_{t+1}^{kj})_+>0} = 1$ then the variation in stocks is positive meaning that we saturated by rows since it implied that the minimum of the margins was not at the jth column but at the jth row.

<u>Case 1</u>: If $\frac{1}{4}$ (1 1 1 1) $\phi((\Delta S_{t+1}^{kj}))$.T $= 1/2$ then we have a table to fill in that is directly computable via some trick of table under margin/column constraints.

<u>Case 2</u>: If $\frac{1}{4}$ (1 1 1 1) $\phi((\Delta S_{t+1}^{kj}))$.T $> 1/2$ then we have three saturated rows and one saturated column meaning that we can easily compute the values that are not set to zero due to migrant minimization and margin constraints.

<u>Case 3</u>: If $\frac{1}{4}$ (1 1 1 1) $\phi((\Delta S_{t+1}^{kj}))$.T $< 1/2$ then we have three saturated rows and one saturated column meaning that we can easily compute the values that are not set to zero due to migrant minimization and margin constraints.

Let us illustrate all the three cases for k = A concerning the migrant minimization procedure.
<u>Illustration of case 1</u>: Example with $\phi((\Delta S_{t+1}^{Aj}))$ = (0,1,0,1).

$$\hat{\mathbf{F}}_{\mathbf{t}}^{\mathbf{A}} := \begin{array}{c} \\ A \\ B \\ C \\ D \\ \sum_j \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & \sum_i \end{array} \\ \begin{pmatrix} S_{t+1}^{A,A} & (\hat{F}_t^A)_{AB} & 0 & (\hat{F}_t^A)_{AD} & S_t^{A,A} \\ 0 & S_t^{A,B} & 0 & 0 & S_t^{A,B} \\ 0 & (\hat{F}_t^A)_{CB} & S_{t+1}^{A,C} & (\hat{F}_t^A)_{CD} & S_t^{A,C} \\ 0 & 0 & 0 & S_t^{A,D} & S_t^{A,D} \\ S_{t+1}^{A,A} & S_{t+1}^{A,B} & S_{t+1}^{A,C} & S_{t+1}^{A,D} & \end{pmatrix} \end{array}$$

<u>Illustration of case 2</u>: Example with $\phi((\Delta S_{t+1}^{Aj}))$ = (0,1,1,1).

$$\hat{\mathbf{F}}_{\mathbf{t}}^{\mathbf{A}} := \begin{array}{c} \\ A \\ B \\ C \\ D \\ \sum_j \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & \sum_i \end{array} \\ \begin{pmatrix} S_{t+1}^{A,A} & (\hat{F}_t^A)_{AB} & (\hat{F}_t^A)_{AC} & (\hat{F}_t^A)_{AD} & S_t^{A,A} \\ 0 & S_t^{A,B} & 0 & 0 & S_t^{A,B} \\ 0 & 0 & S_t^{A,C} & 0 & S_t^{A,C} \\ 0 & 0 & 0 & S_t^{A,D} & S_t^{A,D} \\ S_{t+1}^{A,A} & S_{t+1}^{A,B} & S_{t+1}^{A,C} & S_{t+1}^{A,D} & \end{pmatrix} \end{array}$$

<u>Illustration of case 3</u>: Example with $\phi((\Delta S_{t+1}^{Aj}))$ = (0,1,0,0).

$$\hat{\mathbf{F}}_{\mathbf{t}}^{\mathbf{A}} := \begin{array}{c} \\ A \\ B \\ C \\ D \\ \sum_j \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & \sum_i \end{array} \\ \begin{pmatrix} S_{t+1}^{A,A} & (\hat{F}_t^A)_{AB} & 0 & (\hat{F}_t^A)_{AD} & S_t^{A,A} \\ 0 & S_t^{A,B} & 0 & 0 & S_t^{A,B} \\ 0 & (\hat{F}_t^A)_{CB} & S_{t+1}^{A,C} & 0 & S_t^{A,C} \\ 0 & (\hat{F}_t^A)_{DB} & 0 & S_t^{A,D} & S_{t+1}^{A,D} \\ S_{t+1}^{A,A} & S_{t+1}^{A,B} & S_{t+1}^{A,C} & S_{t+1}^{A,D} & \end{pmatrix} \end{array}$$

Case 1: The mean of the output vector is 1/2. Given $\phi((\Delta S^{kj}_{t+1}))$: 1s will correspond to the column's index of the sub-matrix and 0s will correspond to the row's index of the sub-matrix. The sub-matrix looks like this:

$$\hat{\mathbf{F}}^{\mathbf{k}}_{\mathbf{t|Sub}} = \begin{array}{c} \\ 0_1 \\ 0_2 \\ ColumnMargins \end{array} \begin{array}{c} \begin{array}{ccc} 1_1 & 1_2 & RowMargins \end{array} \\ \begin{pmatrix} & & |\Delta S^{k0_1}_{t+1}| \\ & & |\Delta S^{k0_2}_{t+1}| \\ |\Delta S^{k1_1}_{t+1}| & |\Delta S^{k1_2}_{t+1}| & \end{pmatrix} \end{array}$$

Now, one could remark that the column margins correspond to net outflows from 1_1 and 1_2 of people born in k and that row margins correspond to net inflows for 0_1 and 0_2 of people born in k. Since we are in a closed system and that there is no demographic dynamics requiring any adjustment of our stock tables, we can notice that the sum of the column margins matches with the sum of the row margins of this sub-matrix.

Having a well-defined sub-matrix using difference in stocks as margins and not stocks as margins as it was done for F^k_t, one can use this trick to fill the sub-matrix and therefore the entire birth place specific bilateral migrant flow table as follows:

$$\hat{\mathbf{F}}^{\mathbf{k}}_{\mathbf{t|Sub}} = \begin{array}{c} \\ 0_1 \\ 0_2 \\ ColumnMargins \end{array} \begin{array}{c} \begin{array}{ccc} 1_1 & 1_2 & RowMargins \end{array} \\ \begin{pmatrix} \frac{|\Delta S^{k0_1}_{t+1}|\times|\Delta S^{k1_1}_{t+1}|}{|\Delta S^{k1_1}_{t+1}|+|\Delta S^{k1_2}_{t+1}|} & \frac{|\Delta S^{k0_1}_{t+1}|\times|\Delta S^{k1_2}_{t+1}|}{|\Delta S^{k1_1}_{t+1}|+|\Delta S^{k1_2}_{t+1}|} & |\Delta S^{k0_1}_{t+1}| \\ \frac{|\Delta S^{k0_2}_{t+1}|\times|\Delta S^{k1_1}_{t+1}|}{|\Delta S^{k1_1}_{t+1}|+|\Delta S^{k1_2}_{t+1}|} & \frac{|\Delta S^{k0_2}_{t+1}|\times|\Delta S^{k1_2}_{t+1}|}{|\Delta S^{k1_1}_{t+1}|+|\Delta S^{k1_2}_{t+1}|} & |\Delta S^{k0_2}_{t+1}| \\ |\Delta S^{k1_1}_{t+1}| & |\Delta S^{k1_2}_{t+1}| & \end{pmatrix} \end{array}$$

Like before, for example with $\phi((\Delta S^{Aj}_{t+1})) = (0,1,0,1)$.

$$\hat{\mathbf{F}}^{\mathbf{A}}_{\mathbf{t}} := \begin{array}{c} \\ A \\ B \\ C \\ D \\ \sum_j \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & \sum_i \end{array} \\ \begin{pmatrix} S^{A,A}_{t+1} & \frac{|\Delta S^{AB}_{t+1}|\times|\Delta S^{AA}_{t+1}|}{|\Delta S^{AB}_{t+1}|+|\Delta S^{AD}_{t+1}|} & 0 & \frac{|\Delta S^{AD}_{t+1}|\times|\Delta S^{AA}_{t+1}|}{|\Delta S^{AB}_{t+1}|+|\Delta S^{AD}_{t+1}|} & S^{A,A}_t \\ 0 & S^{A,B}_t & 0 & 0 & S^{A,B}_t \\ 0 & \frac{|\Delta S^{AB}_{t+1}|\times|\Delta S^{AC}_{t+1}|}{|\Delta S^{AB}_{t+1}|+|\Delta S^{AD}_{t+1}|} & S^{A,C}_{t+1} & \frac{|\Delta S^{AD}_{t+1}|\times|\Delta S^{AC}_{t+1}|}{|\Delta S^{AB}_{t+1}|+|\Delta S^{AD}_{t+1}|} & S^{A,C}_t \\ 0 & 0 & 0 & S^{A,D}_t & S^{A,D}_t \\ S^{A,A}_{t+1} & S^{A,B}_{t+1} & S^{A,C}_{t+1} & S^{A,D}_{t+1} & \end{pmatrix} \end{array}$$

This will be easy to implement and to compute in that case since we will simply inject the right values at the right places.

Case 2: The mean of the output vector is greater than 1/2.

In that case, since we have four countries for our scenarios, it means that we have only one place where the stock of foreign-population born in k decreased and for all the others it increased in overall.

Then it means we have three rows saturated with zero values off the diagonal and one column saturated with zero off the diagonal. It means that will have three values to estimate which will correspond to differences in stocks.

Case 3: The mean of the output vector is lesser than 1/2.

In that case, since we still have four countries for our scenarios, it means that we have only one place where the stock of foreign-population born in k increased and for all the others it decreased in overall.

Then it means we have three columns saturated with zero values off the diagonal and one row saturated with zero off the diagonal. It means that will have three values to estimate which will correspond to differences in stocks.

Solution for case 2 and 3: In case 2 and 3, given $\phi((\Delta S^{kj}_{t+1}))$, we will either have a row to fill (the case if zero-rows are dominant) corresponding to the index of the zero-column and vice versa we will have a column to fill (the case if zero-columns are dominant) corresponding to the index of the zero-row.

Then take j as the minority index and J the dominant index. The minority index contains the only 1's index or 0's index depending on the case and the dominant index all the rest.

In case 2:
We proceed likewise: $(\hat{F}^k_t)_{jl} = |\Delta S^{kl}_{t+1}|$ for all l $\in$ J.
In case 3:
We proceed likewise: $(\hat{F}^k_t)_{lj} = |\Delta S^{kl}_{t+1}|$ for every l $\in$ J.

Such that we have $\hat{F}^k_t$ defined for all the existing cases. Indeed, the case where we would have only zeros and only ones as output of $\phi((\Delta S^{kj}_{t+1}))$ does not exist since we are in a closed system.

Existing and Potential Alternatives: Stepwise Ascent (Raymer,Abel and Smith 2007) and Gradient Descent

Concerning the Stepwise Ascent: This is implemented on the R package Migest. We could take a look at it. This method is mentioned as well in Abel (2013).
Concerning the Gradient: Now, we could proceed to Poisson Regression, compute the Gradient, try to bound it to see if B smooth and try to lower bound it as well to see if we can have a condition number.
Setting: Our goal with i $\neq$ j, $\hat{F}^k_{ij} = \mathbf{E}(F^k_{ij}) = \hat{\alpha}^k_i\ \hat{\beta}^k_j$ with $F^k_{ij} \sim P(\hat{\alpha}^k_i \hat{\beta}^k_j)$ where $\hat{\alpha}^k_i$, $\hat{\beta}^k_j$ correspond to MLE estimates.
Since we have migrant minimization setting, let us recall if i = j, $\hat{F}^k_{ij} = \mathbf{E}(F^k_{ij}) = \hat{\alpha}^k_i\ \hat{\beta}^k_j\ \hat{\delta}^k_{ii}$.
Remember, we computed the gradient in annex with the proof of migrant stock data as sufficient statistics. Nevertheless, since there exists already a method used to solve the problem, we could also refer to it with

2) **Computing the error terms**: We might keep the information provided by $\phi((\Delta S^{kj}_{t+1}))$ and rearrange all that to provide a case by case analysis of the error decomposition in terms of transit, emigration and return flows.

The error term is computed like this:
$\hat{F}^{ij}_t$ - F^{ij}_t = $\hat{E}_{ij}$ + $\hat{R}_{ij}$ $\sum_{k\neq i,j} \hat{T}^k_{ij}$ - E_{ij} - R_{ij} - $\sum_{k\neq i,j} T^k_{ij}$ = $(\hat{F}^i)_{ij}$ + $(\hat{F}^j)_{ij}$ + $\sum_{k\neq i,j} (\hat{F}^k)_{ij}$ - E_{ij} - R_{ij} - $\sum_{k\neq i,j} T^k_{ij}$.
Let us analyse this:

$\hat{E}$migration Flows Exploration:

Case 0: $(\phi((\Delta S^{il}_{t+1})))_i = 1$ then $\hat{E}_{ij} = 0$ for every j.

Case 1: $(\phi((\Delta S^{il}_{t+1})))_i = 0$
For all l zero entries of $\phi((\Delta S^{il}_{t+1}))$, since the lth column is saturated with zeros, we have $(\hat{F}^i_t)_{il} = 0$ in other words: if $\Delta S^{ii}_{t+1} < 0$ and $\Delta S^{il}_{t+1} < 0$ then $\hat{E}_{il} = 0$.

Sub-case 1: $\frac{1}{4}$ (1 1 1 1) $\phi((\Delta S^{il}_{t+1})) > 1/2$.
Here, we are in the case where $\Delta S^{ii}_{t+1} < 0$ and for all l $\neq$ i $\Delta S^{il}_{t+1} > 0$ then we have $(\hat{F}^i)_{il} = \Delta S^{il}_{t+1}$. It is the same

as estimating F_t^k via difference method with non negative differences for the stocks in host countries.

<u>Sub-case 2</u>: $\frac{1}{4}$ (1 1 1 1) $\phi((\Delta S_{t+1}^{il})) = 1/2$.
Here, we have that when $\Delta S_{t+1}^{il} > 0$, $(\hat{F}^i)_{il} = \frac{|\Delta S_{t+1}^{ii}||\Delta S_{t+1}^{il}|}{\sum_{z|\Delta S_{t+1}^{iz}>0}|\Delta S_{t+1}^{iz}|}$

<u>Sub-case 3</u>: $\frac{1}{4}$ (1 1 1 1) $\phi((\Delta S_{t+1}^{il})) < 1/2$.
Here, we have that when $\Delta S_{t+1}^{il} > 0$, $(\hat{F}^i)_{il} = |\Delta S_{t+1}^{ii}|$

<u>$\hat{R}$eturn Flows Exploration</u>: By symmetry with the previous reasoning, we have:

<u>Case 0</u>: $(\phi((\Delta S_{t+1}^{jl})))_j = 0$ then $\hat{R}_{ij} = 0$ for every i.

<u>Case 1</u>: $(\phi((\Delta S_{t+1}^{jl})))_j = 1$
For all l 1-entries of $\phi((\Delta S_{t+1}^{jl}))$, since the lth row is saturated with zeros, we have $(\hat{F}_t^j)_{lj} = 0$ in other words: if $\Delta S_{t+1}^{jj} > 0$ and $\Delta S_{t+1}^{jl} > 0$ then $\hat{R}_{lj} = 0$.

<u>Sub-case 1</u>: $\frac{1}{4}$ (1 1 1 1) $\phi((\Delta S_{t+1}^{jl})) < 1/2$.
Here, we are in the case where $\Delta S_{t+1}^{jj} > 0$ and for all l $\neq$ j $\Delta S_{t+1}^{jl} < 0$ then we have $(\hat{F}^j)_{lj} = |\Delta S_{t+1}^{jl}|$. It is the same as estimating F_t^k via difference method with non positive differences for the stocks in host countries.

<u>Sub-case 2</u>: $\frac{1}{4}$ (1 1 1 1) $\phi((\Delta S_{t+1}^{jl})) = 1/2$.
Here, we have that when $\Delta S_{t+1}^{jl} < 0$, $(\hat{F}^j)_{lj} = \frac{|\Delta S_{t+1}^{jj}||\Delta S_{t+1}^{jl}|}{\sum_{z|\Delta S_{t+1}^{jz}<0}|\Delta S_{t+1}^{iz}|}$.

<u>Sub-case 3</u>: $\frac{1}{4}$ (1 1 1 1) $\phi((\Delta S_{t+1}^{jl})) > 1/2$.
Here, we have that when $\Delta S_{t+1}^{jl} < 0$, $(\hat{F}^j)_{lj} = |\Delta S_{t+1}^{jj}|$.

<u>$\hat{T}$ransit Flows Exploration</u>: Consider T_{ij}^k which is at the (i,j) coordinate of $\hat{F}_t^k$. It will be equal to zero if $\Delta S_{t+1}^{ki} \geq 0$ OR if $\Delta S_{t+1}^{kj} \leq 0$.

Now let us look when $\Delta S_{t+1}^{ki} < 0$ AND $\Delta S_{t+1}^{kj} > 0$. Then, we have few sub-cases:

<u>Sub-case 1</u>: For every l $\neq$ i,j $\Delta S_{t+1}^{kl} \geq 0$, then three rows zero-saturated (the jth included) and the ith column zero-saturated. Then $\hat{T}_{ij}^k = |\Delta S_{t+1}^{kj}| = \Delta S_{t+1}^{kj}$.

<u>Sub-case 2</u>: For every l $\neq$ i,j $\Delta S_{t+1}^{kl} \leq 0$, then three columns zero-saturated (the ith included) and the jth row zero-saturated. Then $\hat{T}_{ij}^k = |\Delta S_{t+1}^{ki}| =$ -ΔS_{t+1}^{ki}.

<u>Sub-case 3</u>: There exists a couple (l,l') each respectively $\neq$ i,j — $\Delta S_{t+1}^{kl} \leq 0$ $\Delta S_{t+1}^{kl} \geq 0$. Then, one has:
$(\hat{F}_t^k)_{ij} = \frac{|\Delta S_{t+1}^{ki}||\Delta S_{t+1}^{kj}|}{|\Delta S_{t+1}^{ki}|+|\Delta S_{t+1}^{kl}|}$

Now we are able to automatize the plotting of the decomposition of the error term relative to the estimation of F_{ij}^t via Abel's method with contingency tables.

<u>Plan</u>:

Our plan to decompose the error term relative to $\hat{F}_{ij}^t$ for contingency table method will rely on three steps. First analyse the error term relative to $\hat{E}_{ij}^i$. Second analyse the error term relative to $\hat{R}_{ij}^j$ (we will use symmetry). Third we will analyse the error term relative to, for every k $\neq$ i,j, $\hat{T}_{ij}^k$. All this is possible thanks to our previous explorations.

Now we propose to analyse the error term relative to the estimation of E_{ij}^i in the case of contingency table method.

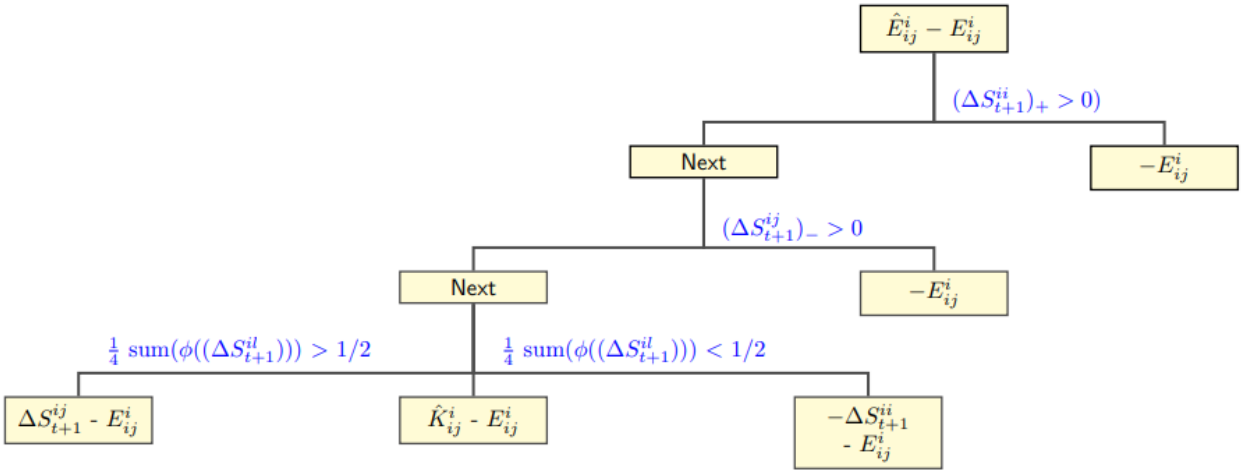


Figure 2: Decision Tree for Error Decomposition of Emigration Flows via Contingency Table Method. Notice that $\hat{K}^i_{ij}$ correspond to sub-case 2 of case 1 of Emigration Flows Exploration.

Analysis of the error term relative to Emigration Flows in the case of contingency table method:

Case 1: $\hat{E}^i_{ij} = 0 \Rightarrow$ -E^i_{ij} as error term decomposition; No estimation, total under-estimation.

Case 2: $\hat{E}^i_{ij} = \Delta S^{ij}_{t+1} \Rightarrow$ -$R^i_{ji} + \sum_{k \neq i,j} T^i_{kj} - T^i_{jk}$ as error term decomposition; Classic Error Decomposition with Tri and RetInv;

Case 3: $\hat{E}^i_{ij}$ = - $\Delta S^{ii}_{t+1} \Rightarrow \sum_{j \neq i} R^i_{ji} - \sum_{j' \neq j} E^i_{ij'} - 2E^i_{ij}$ as error term decomposition; Idea of Multilateral Cross Flow Bias;

Case 4: $\hat{E}^i_{ij} = \frac{|\Delta S^{ii}_{t+1}||\Delta S^{ij}_{t+1}|}{\sum_{k|\Delta S^{ik}_{t+1}>0} |\Delta S^{ik}_{t+1}|} \Rightarrow$ require deeper analysis... some ideas: see next section and below.

Concerning Case 4, we have the total net flows at the denominator and the product of two opposite nodes in terms of signs of the differences in stocks. See figure 3 with k = A and an example of $P\bar{h}iA = 1/2$ i.e. two positive differences in stocks and two negative differences in stocks.

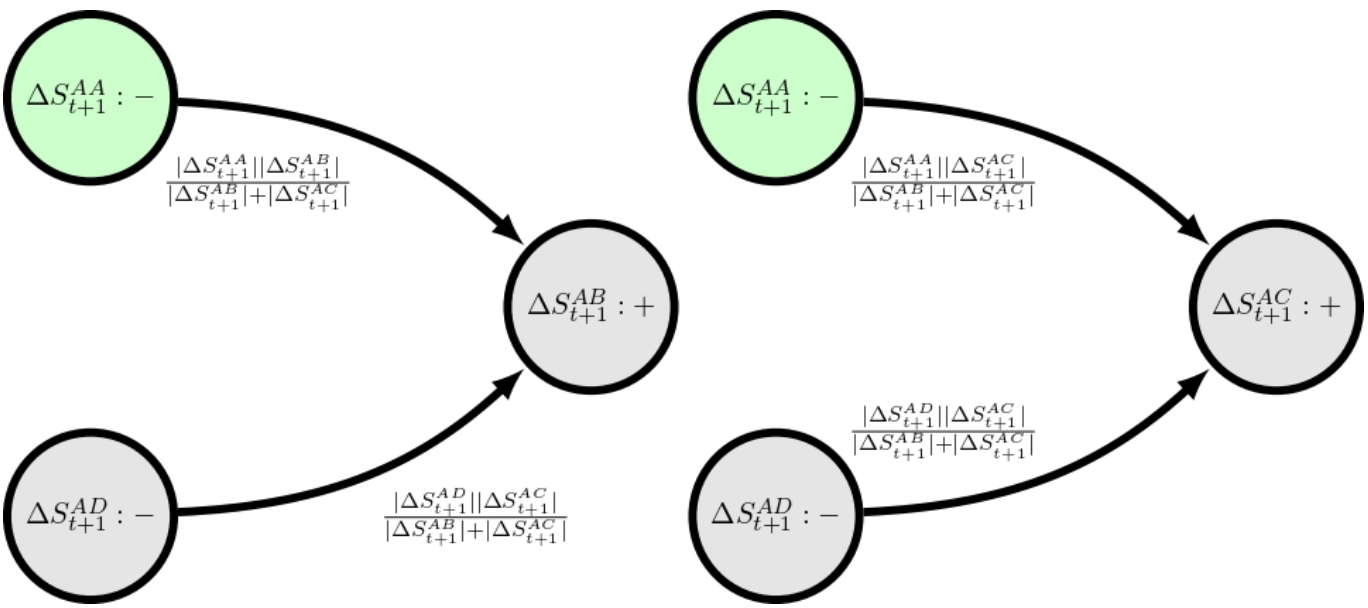


Figure 3: Representation of the estimated emigration/transit flows in sub-cases such that we fill a sub-matrix with equal margins corresponding to absolute value in stock differences here applied to sub-population A with a special case of 2 columns/2 rows of saturated zeros due to migrant minimization assumption

Thanks to this picture, we can then have an idea of what will determine whether there will be higher and lower estimates for these particular flows. In particular, we will need to do ratios or compare differences in stocks of same sign to see which negative nodes will be a greater source of flows and which of the positive nodes will be a greater absorbing country of these flows. This is how we would be able to explain why emigration flows from A to B were under-estimated, this is due to a high value in difference in stocks in absolute value of people born in A and living in D between t and t+1 for instance.

Now let us continue with the analysis of the error term relative to the estimation of R^j_{ij} in the case of contingency table method.

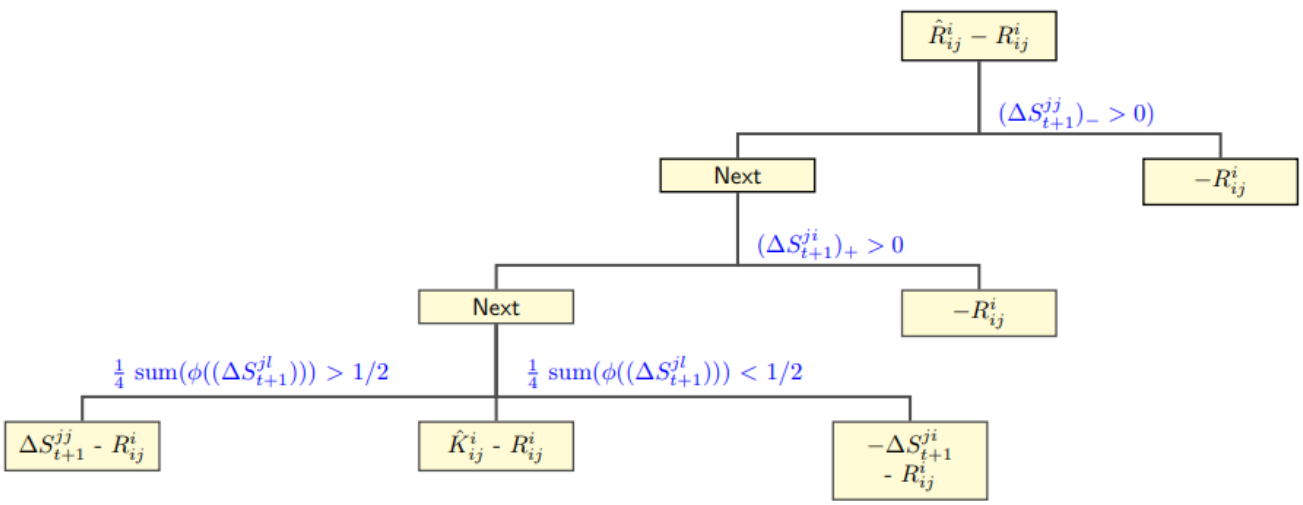


Figure 4: Decision Tree for Error Decomposition of Return Flows via Contingency Table Method. Notice that $\hat{K}^i_{ij}$ correspond to sub-case 2 of case 1 of Return Flows Exploration.

Analysis of the error term relative to Emigration Flows in the case of contingency table method:

Case 1: $\hat{R}^j_{ij} = 0 \Rightarrow$ -R^j_{ij} as error term decomposition; No estimation, total under-estimation.

Case 2: $\hat{R}^j_{ij}$ = -$\Delta S^{ji}_{t+1} \Rightarrow$ -$E^j_{ji} + \sum_{k \neq i,j} T^j_{ik} - T^j_{ki}$ as error term decomposition; Classic Error Decomposition with Trj and EmigInv;

Case 3: $\hat{R}^j_{ij} = \Delta S^{jj}_{t+1} \Rightarrow \sum_{j \neq i} E^j_{ji} - \sum_{j' \neq j} R^j_{ij'} - 2R^j_{ij}$ as error term decomposition; Idea of Multilateral Cross Flow Bias;

Case 4: $\hat{R}^j_{ij} = \frac{|\Delta S^{jj}_{t+1}||\Delta S^{ji}_{t+1}|}{\sum_{k|\Delta S^{ik}_{t+1}>0} |\Delta S^{ik}_{t+1}|} \Rightarrow$ same idea as developed for emigration flows estimation.

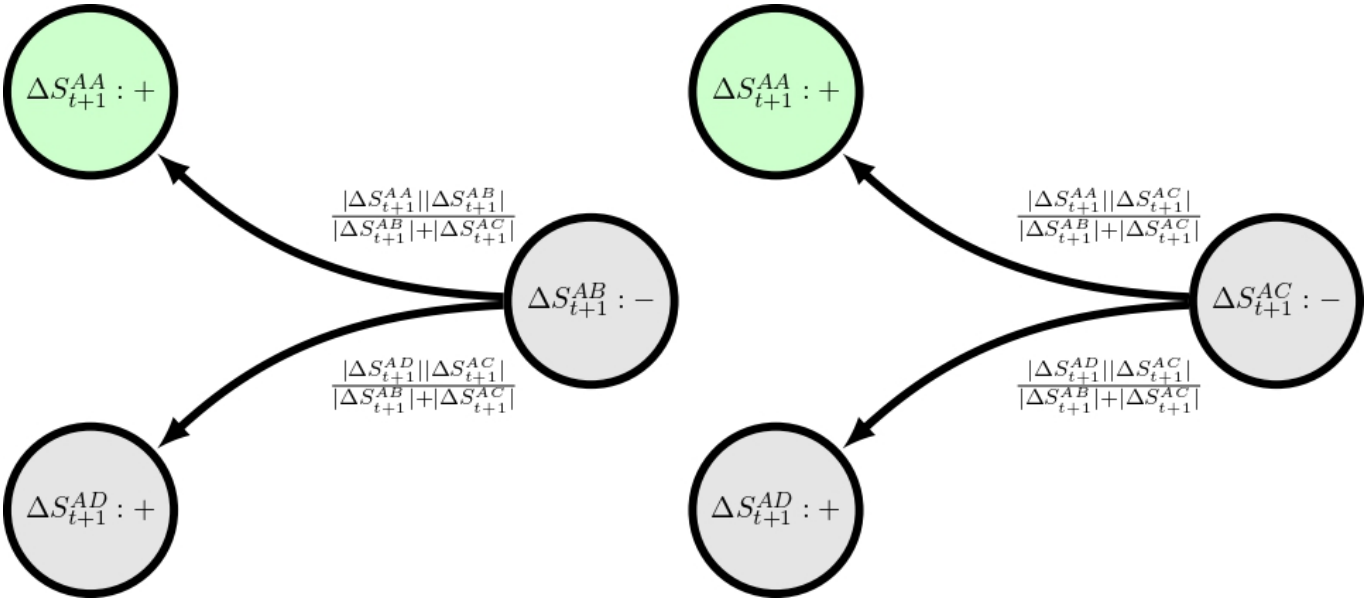


Figure 5: Representation of the estimated return/transit flows in sub-cases such that we fill a sub-matrix with equal margins corresponding to absolute value in stock differences here applied to sub-population A with a special case of 2 columns/2 rows of saturated zeros due to migrant minimization assumption

Now we propose to analyse the error term relative to the estimation of T^k_{ij} for every k in the case of contingency table method.

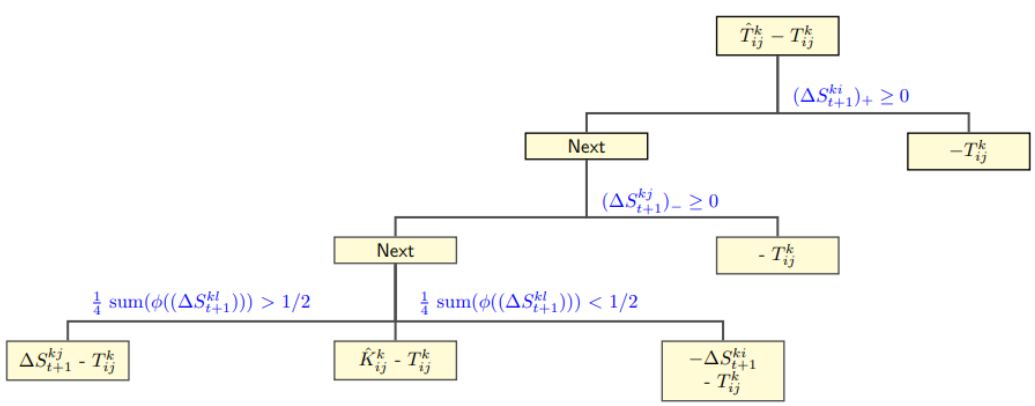


Figure 6: Decision Tree for Error Decomposition of Transit Flows via Contingency Table Method. Notice that $\hat{K}^i_{ij}$ correspond to sub-case 3 of case 1 of Transit Flows Exploration.

Analysis of the error term relative to Transit Flows in the case of contingency table method:

Case 1: $\hat{T}^k_{ij} = 0 \Rightarrow$ -T^k_{ij} as error term decomposition; No estimation, total under-estimation.

Case 2: $\hat{T}^k_{ij} = \Delta S^{kj}_{t+1} \Rightarrow E^k_{kj}$ - R^k_{jk} - T^k_{ji} + $\sum_{l \neq k,i,j}$ (T^k_{lj} - T^k_{jl})

Case 3: $\hat{T}^k_{ij} =$ -$\Delta S^{ki}_{t+1} \Rightarrow$ -E^k_{ki} + R^k_{ik} - T^k_{ji} - $\sum_{l \neq k,i,j}$ (T^k_{li} - T^k_{il})

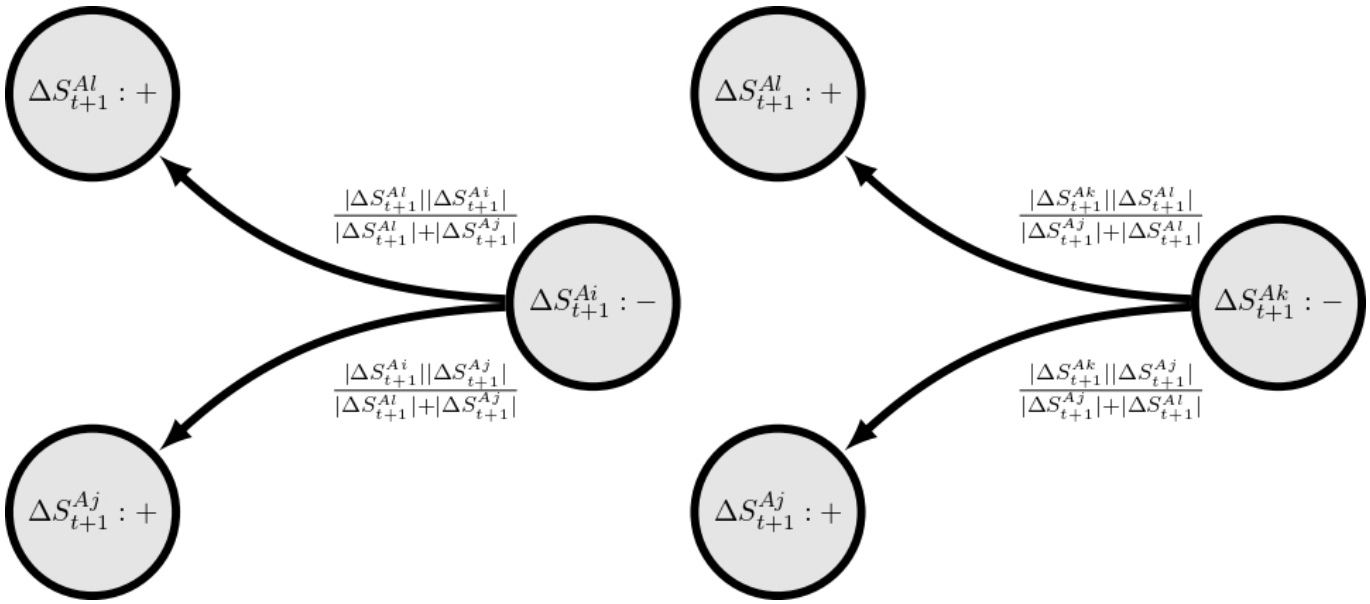


Figure 7: Representation of the estimated transit flows and (emigration OR return flows, depending on the sign of the difference on the diagonal) in sub-cases such that we fill a sub-matrix with equal margins corresponding to absolute value in stock differences here applied to sub-population A with a special case of 2 columns/2 rows of saturated zeros due to migrant minimization assumption

1.7.1 Analysis of the estimated values with contingency table method

We would like to especially understand this quantity and the logic underlying the decision to estimate transit, emigration or return as such a quantity:

For example, with $\hat{R}^j_{ij}$ in case 1, sub-case 2 of return exploration:
$\frac{|\Delta S^{jj}_{t+1}||\Delta S^{jl}_{t+1}|}{\sum_{z|\Delta S^{jz}_{t+1}<0}|\Delta S^{iz}_{t+1}|}$

Our intuition is to represent positive nodes and negative nodes and to see how positive nodes are connected to negative nodes.

Then the interpretation would be how return flows are connected to positive differences on the diagonal of the migrant stock table and how this depend on how other flows like transits can explain as well positive differences on the diagonal.

We would like to study this for later and offer better interpretation and therefore better analysis of the resulting error term to understand the causes underlying the bias. We thought about ratios between stock differences of same signs. For example, in the case of the estimation of emigration flows and in the sub-cases where there are negative differences on the diagonal as well as positive differences for the foreign migrant stock of interest, in the case of as many positive as negative nodes, then we propose to analyse how the emigration flows are under-estimated via a ratio of foreign stocks in the country of destination of the emigration flow and in the country where we also observe between t and t+1 a positive variation in stocks. Since we already have under-estimation in the estimation of emigration flows, this would increase this under-estimation and we would use this ratio to explain it. Of course, we would also explain how cross-flows conditional on place of birth (effect of return inverse) affects the level of estimation of emigration flows in the negative way.

References

Abel, G. J. (2013). Estimating global migration flow tables using place of birth data. *Demographic Research*, 28:505–546.

Abel, G. J., Brottrager, M., Cuaresma, J. C., and Muttarak, R. (2019). Climate, conflict and forced migration. *Global environmental change*, 54:239–249.

Azose, J. J. and Raftery, A. E. (2019). Estimation of emigration, return migration, and transit migration between all pairs of countries. *Proceedings of the National Academy of Sciences*, 116(1):116–122.

Behrends, E. (2000). *Introduction to Markov chains*, volume 228. Springer.

Beine, M., Docquier, F., and Özden, Ç. (2011). Diasporas. *Journal of Development Economics*, 95(1):30–41.

Beine, M. and Parsons, C. (2015). Climatic factors as determinants of international migration. *The Scandinavian Journal of Economics*, 117(2):723–767.

Byron, M. and Condon, S. (1996). A comparative study of caribbean return migration from britain and france: towards a context-dependent explanation. *Transactions of the Institute of British Geographers*, pages 91–104.

Draženović, I., Kunovac, M., and Pripužić, D. (2018). Dynamics and determinants of emigration: the case of croatia and the experience of new eu member states. *Public Sector Economics*, 42(4):415–447.

Hagan, J. M. and Thomas Wassink, J. (2020). Return migration around the world: An integrated agenda for future research. *Annual Review of Sociology*, 46:533–552.

Markov, A. A. (1906). Extension of the law of large numbers to dependent quantities. *Izv. Fiz.-Matem. Obsch. Kazan Univ.(2nd Ser)*, 15(1):135–156.